

Jingyi Weng

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EDUCATION

Capital Normal University

Beijing, China

Computer Science and Technology

Sept. 2022 - Jun. 2026 (Expected)

- **Average Score:** 89.98 / 100 (Ranking: 6/64)
- **Core Course:** Linear Algebra(96), Probability Theory and Statistics(87), Data Structures and Algorithms(90), Graph Theory(89), Principles of Artificial Intelligence(89), Digital Image Processing(93)

PUBLICATIONS

- [1] Xin, J., **Weng, J.**, Zhao, J., Ding, H., Chen, N., Shao, Z., et al. (2025). ODQ-UNet: A quaternion-based U-Net integrating OD and RGB space for nuclear segmentation. *Proc. IEEE Int. Conf. Bioinform. Biomed.* (Accepted).
- [2] Xin, J., **Weng, J.**, Zhao, J., Ding, H., Luo, O., Meng, F., et al. (2025). SRL-UNet: An improved residual U-Net with 2D-selective-scan for nuclear segmentation. *Proc. Int. Conf. Intell. Comput.*, pp. 512–523.
- [3] Xin, J., **Weng, J.**, Zhao, J., & Ding, H. (2025). SSCA-UNet: Improved U-Net for nuclear segmentation by 2D selective-scan and channel attention. *Proc. SPIE Int. Conf. Digit. Image Process.*, vol. 13709, pp. 315–323.

RESEARCH EXPERIENCE

Research on Semantic Segmentation of Microscopic Coal Maceral Image

Undergraduate Thesis

Oct. 2025 - Present

- Develop a network backbone based on UNet++ to adapt to multi-scale object segmentation, and Mamba modules are introduced in the encoding stage to extract rich global contextual information.
- Explore the adaptation of DenseCLIP for microscopic coal maceral segmentation by refining text prompts and improving text-pixel alignment to better incorporate descriptive textual information.

Research on Deep Learning for Histopathological Image Segmentation in Colorectal Cancer

Core Collaborator

Oct. 2024 - Aug. 2025

- Introduced the Optical Density (OD) space within the ODQ-UNet framework, where the blue channel of the OD space is encoded as the real quaternion component to enhance contrast and achieve stain-invariant representation.
- Proposed the Logarithmic Dice-BCE (LDB) loss function by integrating a logarithmic transformation of Dice loss with binary cross-entropy, dynamically adjusting class-specific weights to improve sensitivity to subtle overlaps and mitigate the effects of class imbalance.
- Integrated a 2D-Selective Scanning Module after the first three convolutional layers in the encoder of SSCA-UNet and SRL-UNet, enhancing spatial context capture and improving feature representation for segmentation tasks.
- Introduced Grad-CAM into the segmentation framework to visualize models' attention and validate its focus on nuclear structures, the proposed model achieved a 3.39% higher Dice score compared with other advanced models.
- Presented orally on SSCA-UNet: improved U-Net for nuclear segmentation by 2D selective-scan and channel attention at the 17th International Conference on Digital Image Processing (ICDIP 2025).

Research on Airway Tree Segmentation in Medical Imaging

Core Collaborator

Jan. 2024 - Sept. 2024

- Conducted large-scale comparative experiments across five open-source state-of-the-art models (such as Swin-UNet and UMamba) and two datasets on airway tree segmentation tasks, systematically evaluating model performance, analyzing robustness, and developing in-depth understanding of deep learning.
- Contributed to the design of an airway tree segmentation system with Python, developing core modules such as segmentation algorithms, 3D visualization and angle measurement, with software copyright successfully obtained.

Project on Embedded Control and PID Optimization for Autonomous Line-Following Vehicles

Project Leader Sept. 2022 - Jan. 2023

- Learned key aspects of embedded line-following vehicles, including sensor signal processing, real-time control strategies, system calibration, and integration of core hardware modules, and implemented and deployed algorithms on a microcontroller.
- Optimized adaptive PID control algorithms on a K60 microcontroller-based vehicle for grayscale and electromagnetic line-following tasks, integrating sensor feedback for trajectory correction and differential motor control for steering, performing real-time adjustments and performance tuning, enabling higher speed while accurately handling complex track elements, including intersections, curves, and ramps.

HONORS & AWARDS

Academic Awards

- Second Prize Scholarship of Capital Normal University 2025, 2024, 2023
- Research Innovation Individual Award of Capital Normal University 2025, 2024

Competition Honors

- 1st Prize, National College Student Statistical Modeling Competition, Beijing Division 2025
- 3rd Prize, Chinese Collegiate Computing Competition, National Level 2024
- Honorable Mention, Mathematical Contest in Modeling (MCM/ICM), International Level 2024
- 3rd Prize, Challenge Cup National College Student Business Plan Competition, Beijing Division 2024
- 2nd Prize, Contemporary Undergraduate Mathematical Contest in Modeling, Beijing Division 2024
- 1st Prize, Contemporary Undergraduate Mathematical Contest in Modeling, Beijing Division 2023

TECHNICAL PROFICIENCIES

Programming Skills: Python, C, C++, HTML, Matlab, verilog, JavaScript

Development Tools: PyTorch, Linux, LaTeX

Language: Chinese Mandarin, English (TOEFL: 97; R29/L24/S20/W24)